Qatar University

College of Engineering

Department of Computer Science and Engineering

Internship Report
At National Planning Council

Digitizing a Paper-Based Survey

[redacted]

Mentor [redacted]

2025

This project report is submitted to the Department of Computer Science and Engineering of Qatar University in partial fulfillment of the requirements of the Practical Training course.

Abstract

During my internship at the National Planning Council (NPC), I was introduced to a variety of tools used within the organization, including Oracle APEX, Power BI, .NET, SQL Server, and Python. Each week focused on learning a new technology and then through hands-on tasks we applied what we learned during the week. During the internship I developed proficiency in data analysis, data visualization, low-code application development, and full-stack project implementation. In the final project, I integrated these technologies to develop a functional application that combined a .NET front end, a SQL Server back end, and Power BI reporting. This experience significantly enhanced my technical capabilities and provided practical exposure to industry relevant tools.

Acknowledgment

I would like to express my thanks to Ms. [redacted] my main mentor, for her continuous guidance and support and training throughout the internship, and also to Mr. Wasim Ashraf and Mr. Jameer Ummer for their valuable training sessions and for generously sharing their useful insights. And thank you to Dr. Saleh Alhazbi, for his academic guidance, encouragement, and feedback.

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1. Background of the organization

The Statistical Systems Department is a department at The National Planning Council in Qatar. The department specializes in automating statistical processes starting from data collection, processing the data, making valuable conclusions, publishing and updating. They also upgrade statistics and contribute in the statisticians' decision-making. Some of the department's main responsibilities and activities are to:

- Study, propose, and implement IT projects and solutions to automate statistical processes
- They manage, update and maintain statistical databases and monitor them
- They supply NPC with statistical data and create applications that allow other departments to extract their own data
- They manage database connections with other government agencies

The Statistical Department's Organizational Structure

<u>Four main sections</u>

- Advanced Product Innovation and Design
- National Data and statistical platform
- Statistical Applications and Tools
- Geographical Information Systems

Main tools and technologies used by the Department

- Oracle and Microsoft databases are used to store and compile data for statistical systems and electronic surveys
- Oracle BI and Power BI are used to create interactive reports for various statistics and electronic systems
- Data Collection Systems: The department develops and uses online surveys and opinion polls, which are launched on tablets and a website using the latest technology.

2. Internship learning objectives

<u>Activity Objectives</u>

- Complete hands-on data analysis exercises using Python
- Develop interactive dashboards using Power BI to visualize and interpret data
- Build a mini inventory app using Oracle APEX
- Design a basic relational database in Microsoft SQL Server
- Participate in a full-stack team project by combining .NET or Oracle APEX for the front-end, SQL Server or Oracle DB for the backend, and Power BI or Python for data reporting.
- Deliver a final project presentation and demo

<u>Growth Objectives</u>

- To develop a foundational understanding of web application development using the .NET framework and C#.
- To gain practical knowledge of full-stack application development by integrating front-end, back-end.
- To improve my proficiency in using Power BI for building interactive dashboards and performing data analysis using DAX expressions.

- To understand how to connect and utilize multiple technologies such as .NET, SQL Server, and Power BI.
- To develop effective time management skills by planning and executing weekly tasks in accordance with the structured training schedule.

Significance of the Objectives to Me:

These objectives provide a structured and practical foundation for my career. By actively working on these, I will gain hands-on experience with key technologies like .NET, SQL Server, and Power BI. This will not only bridge the gap between academic theory and real-world application but also help me develop essential professional skills such as time management, teamwork, and effective communication, all of which are crucial for my personal and professional growth.

Connection to the Organization's Mission:

I am eager to apply my learning to the organization's mission of using technology to create innovative data solutions. By contributing to the development and maintenance of these systems, I can directly support the organization's core operations and enhance its ability to gather valuable business intelligence.

3. Internship Experience

Challenges faced:

Throughout the internship, there were points in the journey where it felt like a lot of new information at short periods of time before moving to the next. To cope, I would go back home and try to watch videos, practice more and search to get a better grasp on the ideas before we move onto the next.

Week 1: Data Analysis and Visualization with Python

In the first week I gained an understanding of core Python data structures essential for data manipulation, Series, DataFrame and Libraries such as Numpy and Pandas. I acquired practical skills in preparing and transforming data for analysis by doing Data Cleaning, Filtering, Updating and then creating meaningful Visualizations

Our Task for Week one:

To choose a dataset from the Qatar Open Data Portal related to a specific theme then apply what we learned.

Dataset 1: I decided to go with the dataset explores registered deaths among age group, and marital status since it had a good amount of data to work with.

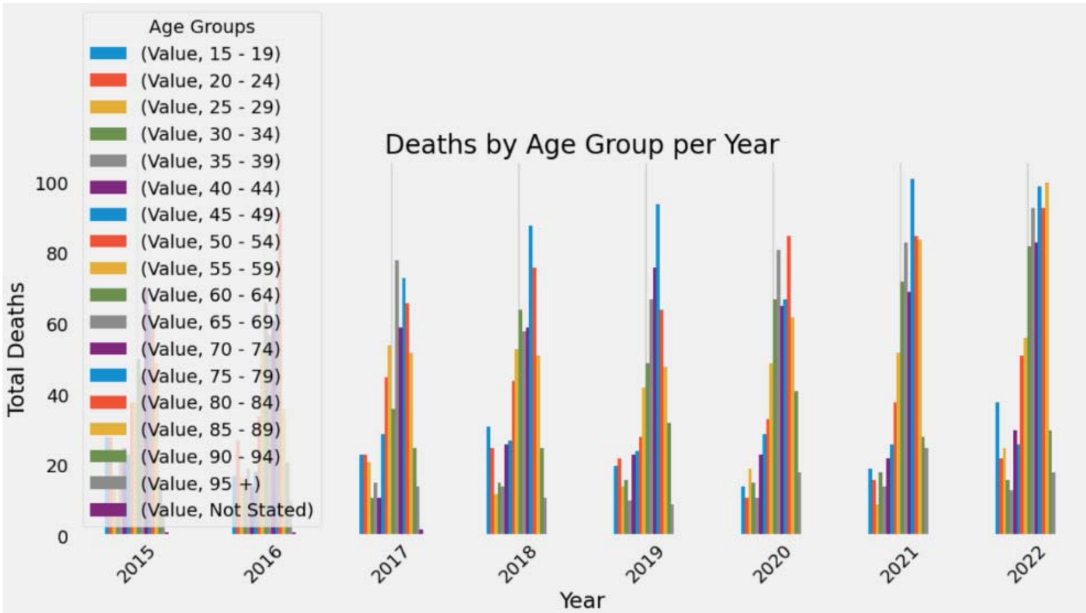


Figure 1: Grouped deaths by year and drew a bar chart to compare each age group, each year

Note: I do realize now that a line graph would have been more suitable for such a chronological visualization.

Dataset 2: This dataset focuses on the causes of death (e.g., Diabetes, heart disease, etc.)

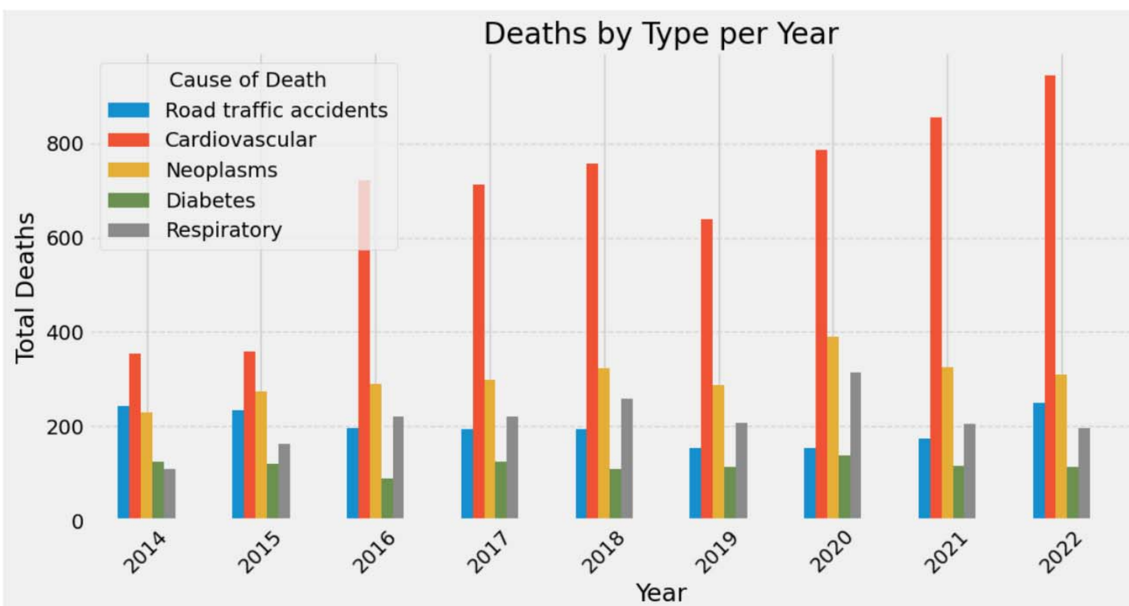


Figure 2: Grouped multiple age groups together to minimize cluster

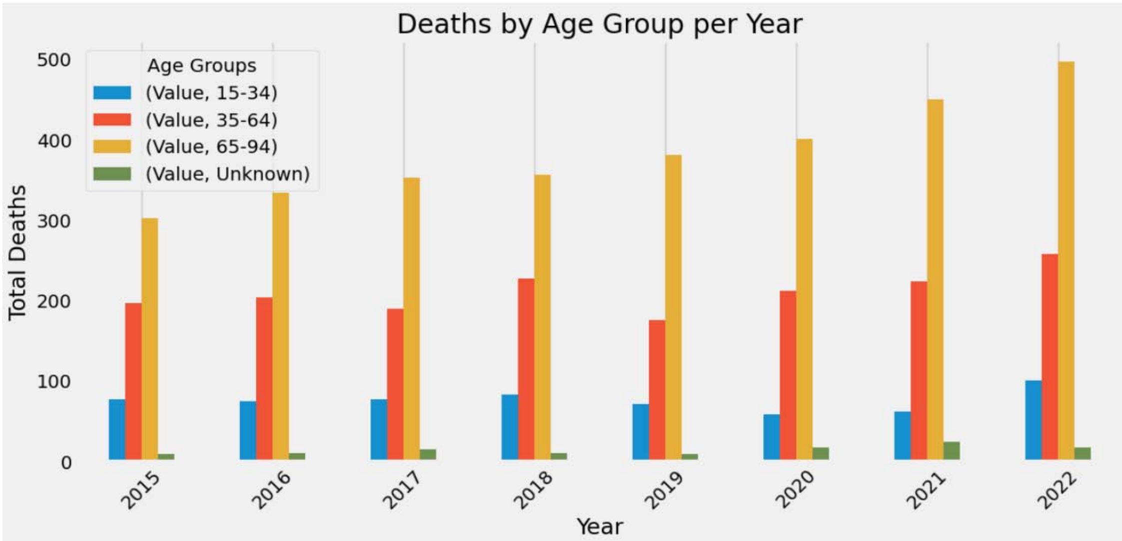


Figure 3: I visualized these to understand which diseases contribute most to mortality and how their prevalence changes over time and to try and connect both datasets by association.

Key Observations

- A consistent rise in deaths in the elderly age group (65–94) from 2020 to 2022.
- Diseases like cancer and heart disease appear to be dominant causes that also have seen a consistent rise during those years, potentially explaining the previous point, by association and causality.

Week 2: Interactive Dashboard Creation with Power BI

The second week transitioned into focusing on creating interactive dashboards using Power BI Desktop. I enhanced my understanding in Data Loading, Data Modeling, and DAX Measures, and how to create an Interactive Dashboard Design.

Our Task for Week two:

To create an interactive Dashboard applying what we learned.

Page 1: The dashboard I created consisted of two pages based on a Book Store data, first page has 3 cards, contained measures I add using DAX, Total Sales amount, Total Quantity amount and Number of Unique books sold. I also added a bar graph showing The Total Sales by each Genre. I also added a Slicer that will filter the data based on book prices and 2 Bookmarks that will show All Data or Top Genre's Data and Finally a button that will take you to the second page, Book Details.

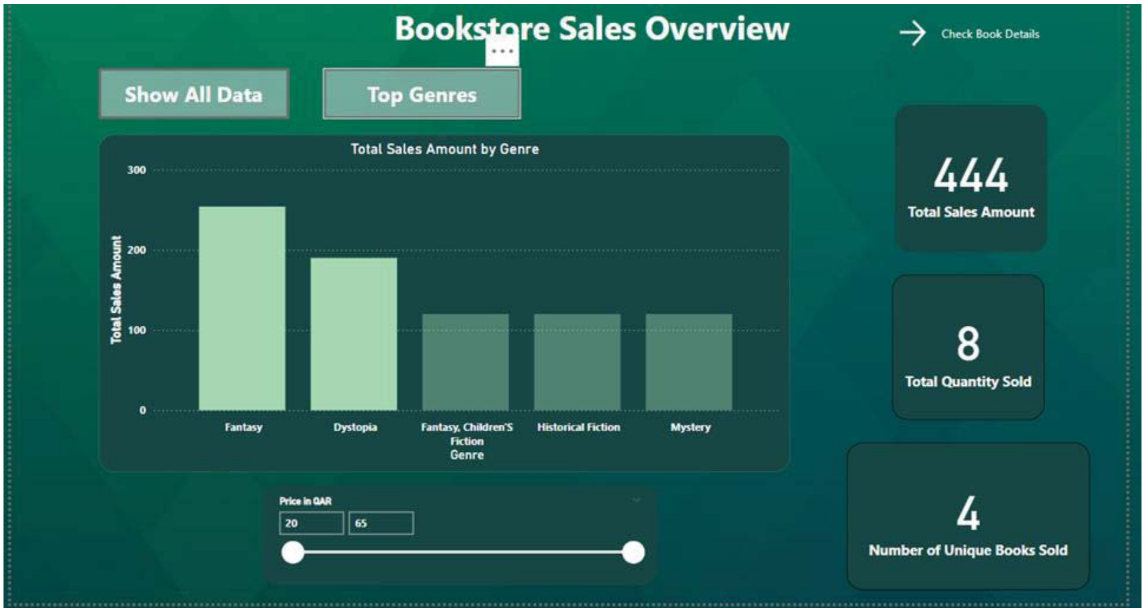


Figure 4: Page 1, Overview page

Page 2: Second page has a detailed table containing information about the Book, Author, Genre, Publisher and DAX measures, Total Quantity sold and Total sales amount. It also contains a dropdown slicer that filters in the data based on Publisher as well as a button that takes you to the Overview page.

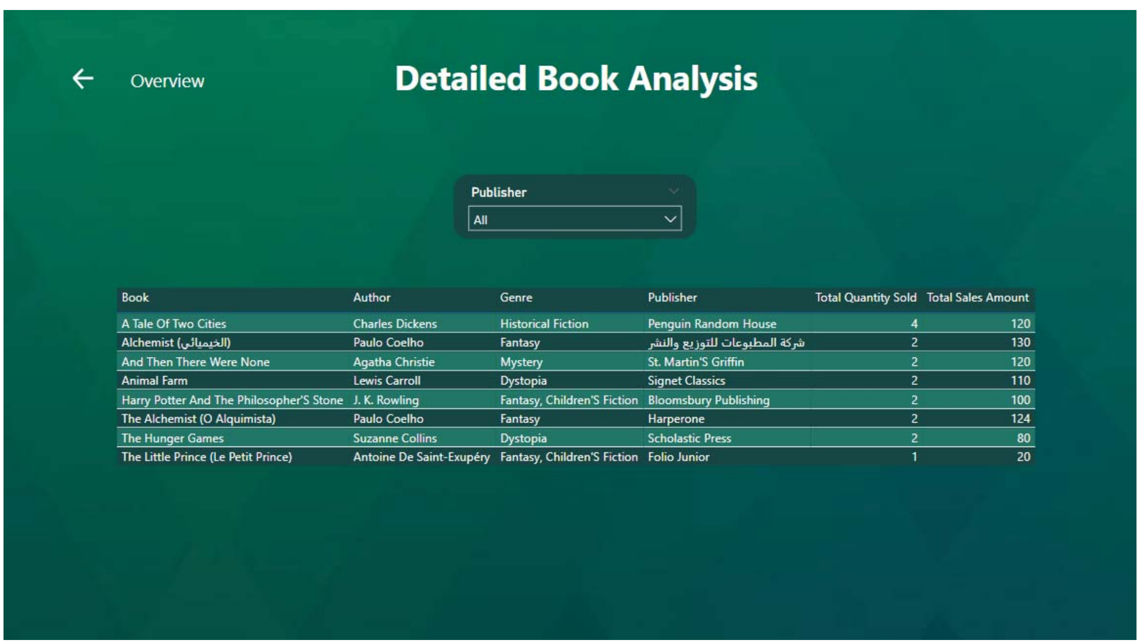


Figure 5: Page 2, Analysis page

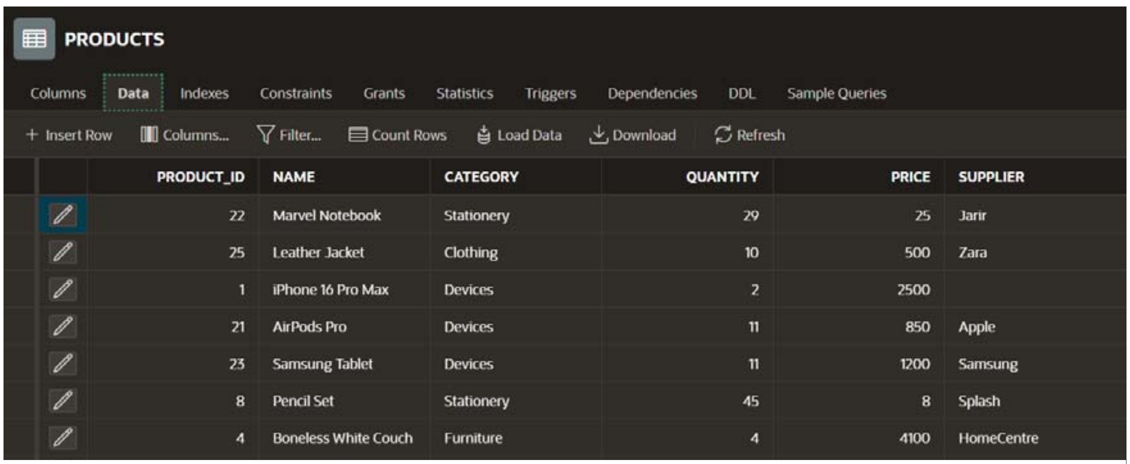
Week 3: Simple Application Development with Oracle APEX

In the third week I got introduced to the basics of web application development using Oracle APEX, a low-code development platform. I gained an understanding of Workspace navigation, Page creation, Writing SQL and more.

Our Task for Week three:

To create a simple app implementing what we learned.

For my app I created an Inventory Management app by creating a Products Table and a Transactions Table. The app contained multiple pages, list of available products and a form to log in new ones, and the ability to edit and delete. The same goes for transactions. Then I created a dashboard with 2 cards, a bar chart and line graph necessary for inventory management.



PRODUCT_ID	NAME	CATEGORY	QUANTITY	PRICE	SUPPLIER
22	Marvel Notebook	Stationery	29	25	Jarir
25	Leather Jacket	Clothing	10	500	Zara
1	iPhone 16 Pro Max	Devices	2	2500	
21	AirPods Pro	Devices	11	850	Apple
23	Samsung Tablet	Devices	11	1200	Samsung
8	Pencil Set	Stationery	45	8	Splash
4	Boneless White Couch	Furniture	4	4100	HomeCentre

Figure 6: Products Table [similarly one for Transactions]

Figure 7: Add Product page

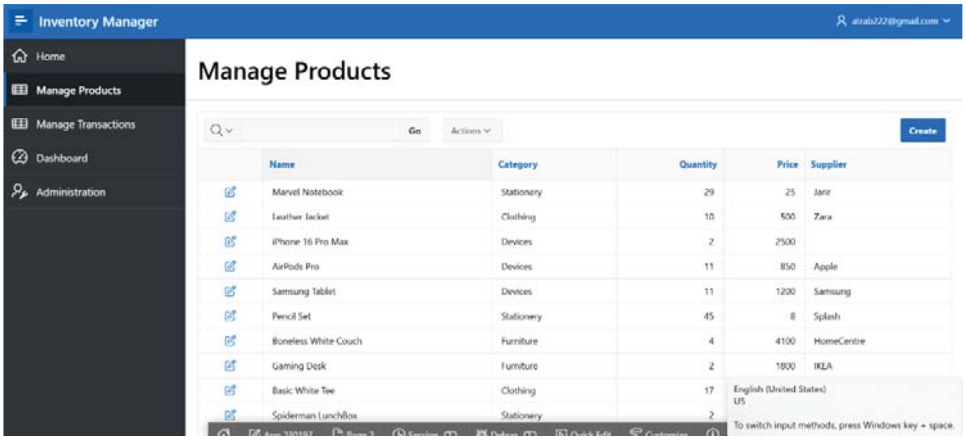


Figure 8: Page to manage Products such as edit, delete and filter [similarly one for Transactions]

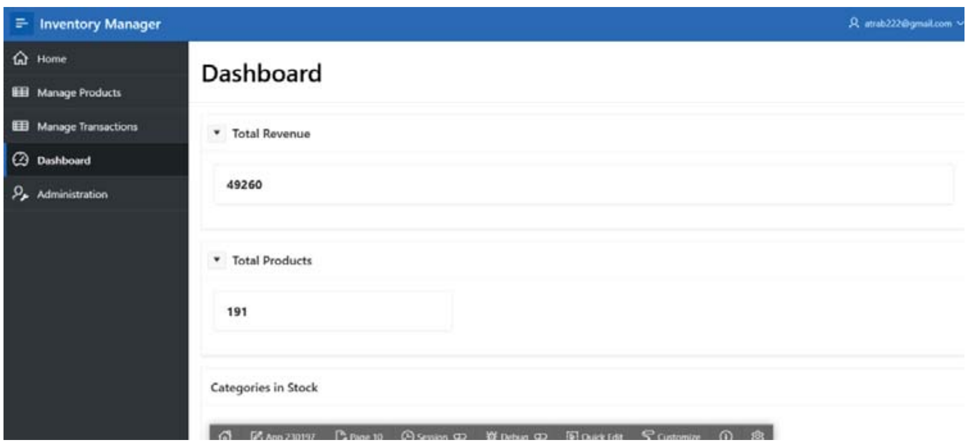


Figure 9: Dashboard page

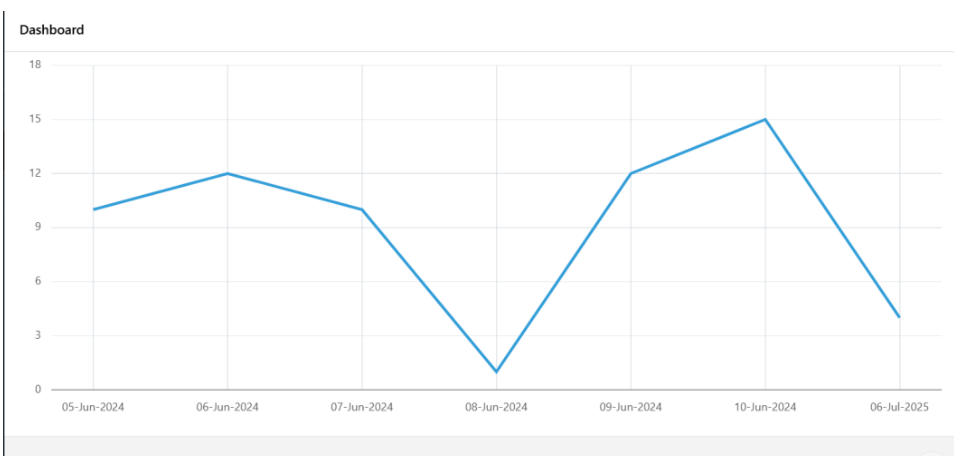


Figure 8: Dashboard graph of the orders in chronological order

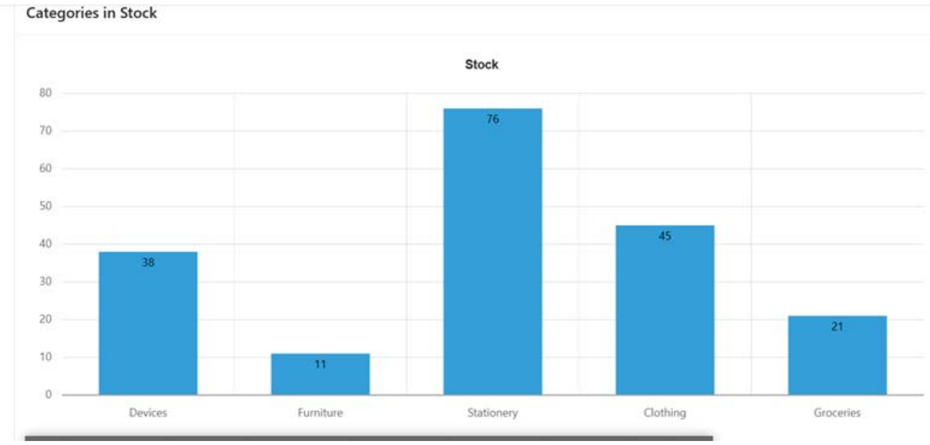


Figure 9: Dashboard bar graph of the categories in stock

Week 4: Simple Development with .NET

Firstly, we took a session on OOP using C# in visual studio, we created some classes, practiced inheritance, polymorphism and encapsulation. We then moved on to learning about .NET Web APIs, why they are necessary, how to set them up, and how to test endpoints for CRUD operations. This helped us understand how backend systems expose data and functionality to client applications.

Our task for week four:
We were asked to choose one of several scenarios and implement it using OOP principles, I selected the Online Bookstore System, where we designed a system to manage physical books and e-books. I created a Book.cs class, with an ID, Name, Author and stock and two abstract methods, calculate Discount and Display Details. Then I created two classes that extended the class Books.cs, Ebook.cs and PhysicalBooks.cs that uniquely implemented the abstract methods. And finally I had the Program.cs where I instantiated two objects of both classes and called the methods.

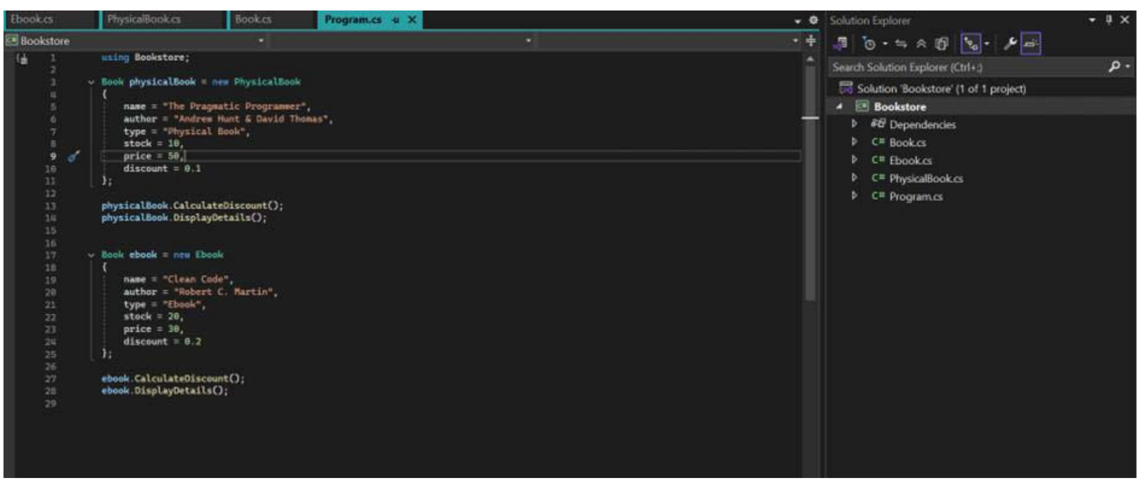


Figure 10: Structure of the task

Week 5: SQL Server

Finally, we explored SQL Server and worked on designing and implementing relational databases using SQL Server Management Studio (SSMS). We were made familiar with the interface and learned how to create tables with primary and foreign keys, define relationships etc.

Our task for week five:
Creating a backend database for an online bookstore system, populating it with sample data, and writing queries, stored procedures. Therefore, 6 Tables were created, Authors, Books, Categories, Customers, OrderItems, and Orders where a one-to-many relationship exists where a single Customer can place multiple Orders. Each of these Orders can contain multiple OrderItems, with each item representing a specific book. The books themselves are organized in a one-to-many structure, where an Author can write many Books, and each book is assigned to a specific Category.

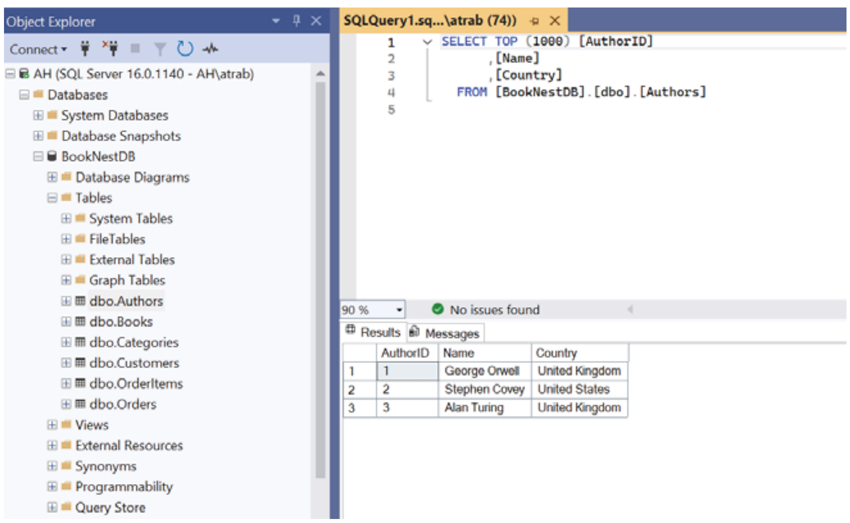


Figure 11: Tables created and example of data

```sql
ALTER PROCEDURE [dbo].[GetCustomerPurchaseHistory]
    @CustomerID INT
AS
BEGIN
  SELECT B.Title FROM Books B INNER JOIN OrderItems OT ON B.BookID=OT.BookID INNER JOIN Orders O ON O.OrderID=OT.OrderID
  WHERE O.CustomerID =  @CustomerID
END
```

Figure 12: Procedure created

Week 6 and 7: Full stack end of internship project
Towards the end of our internship, we were tasked digitize a paper based survey into a web-based system where data is collected, stored, analyzed, and visualized in a dashboard. Using the skills we've learned (Power BI, Python, Oracle APEX, .NET, SQL Server/Oracle DB). I decided to go with .NET for front end and SQL Server for back end and Power Bi for visualizations. We were first required to choose a survey from the Doha Open Portal, then create a Gantt chart then build the Survey then setup the Database then create a Dashboard and finally deliver it in a presentation.

Challenges faced:
During the Final Project I decided to go with .NET instead of Oracle for the challenge and to take the chance to learn more about it considering I have background on web development. However there were struggles that came with that with the tight deadline, my foundational knowledge and I had to refresh my web development knowledge, watch videos and search to be able to successfully come out with a satisfactory and complete project.

1. Gantt chart

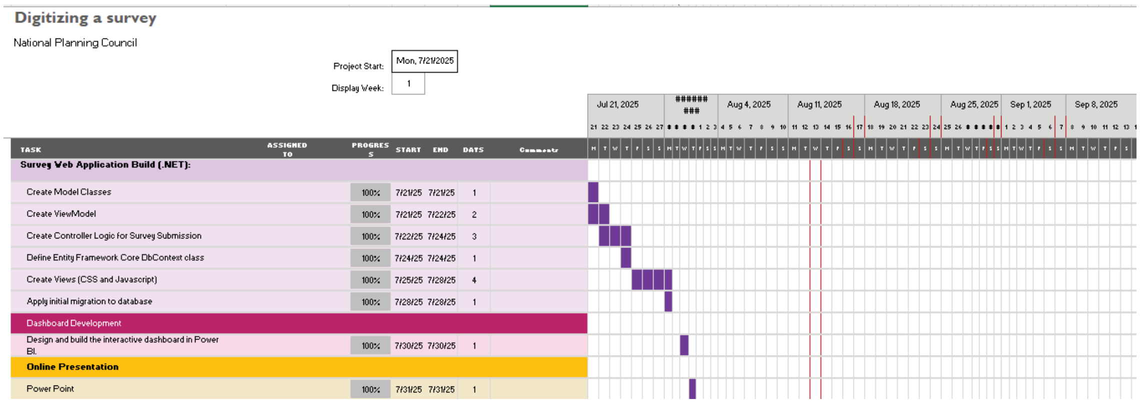
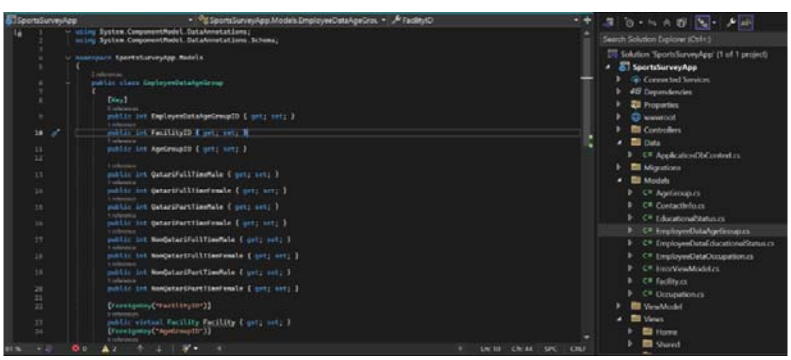


Figure 13: Gantt chart

2. Building survey using .NET

The App was built using:
Models: Database blueprint. What data looks like in the database.
ViewModels: Where data is selected, combined and calculated for the screen
Controller: Gets requests, talks to Model, sends data to View. (a connecter basically)
View: The web page HTML



Database Context: The data connector to Link the app to SQL Server

Figure 14: Overview of the structure

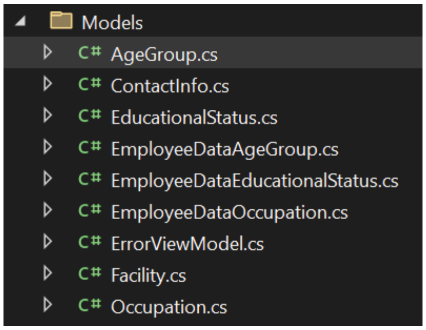


Figure 15: Models created

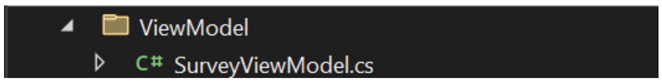


Figure 16: Viewmodel Created

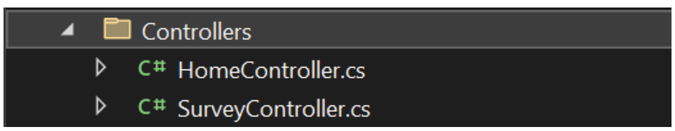


Figure 17: Controller created

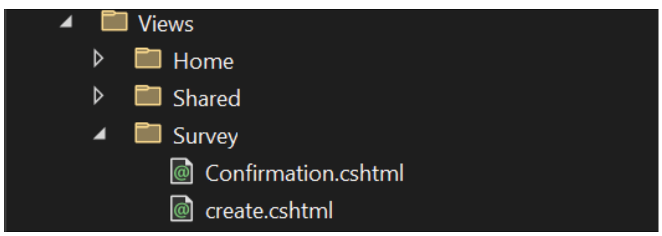
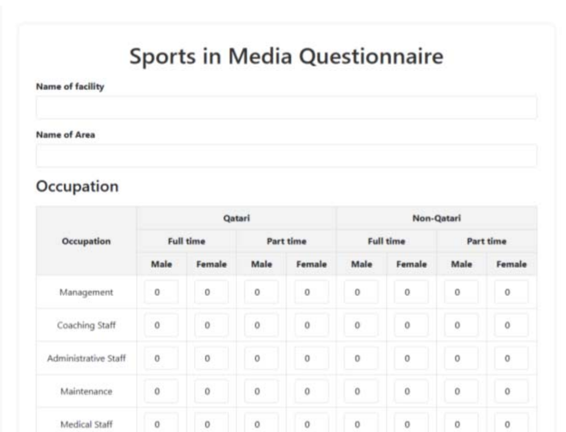


Figure 18: Views created

Figure 19: DB Context created



Sports in Media Questionnaire

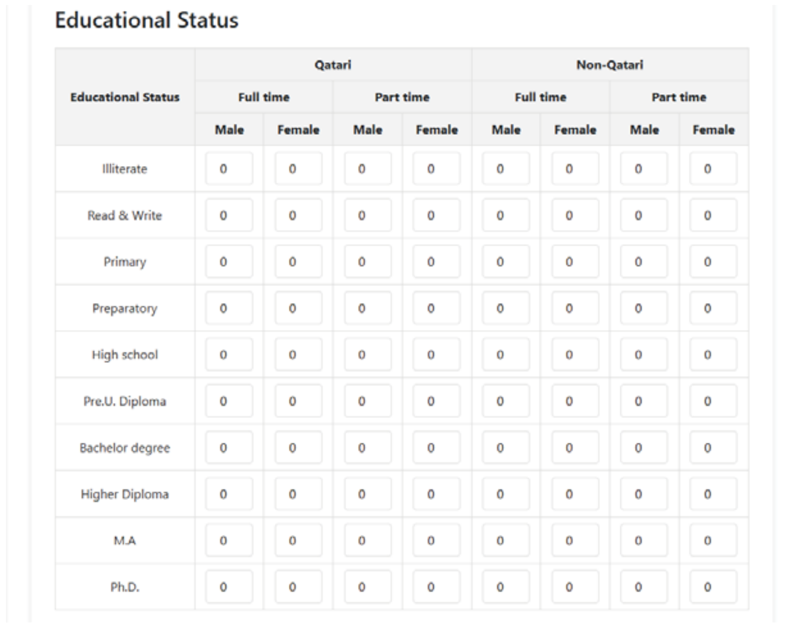
Name of facility

Name of Area

Occupation

Occupation	Qatari				Non-Qatari			
	Full time		Part time		Full time		Part time	
	Male	Female	Male	Female	Male	Female	Male	Female
Management	0	0	0	0	0	0	0	0
Coaching Staff	0	0	0	0	0	0	0	0
Administrative Staff	0	0	0	0	0	0	0	0
Maintenance	0	0	0	0	0	0	0	0
Medical Staff	0	0	0	0	0	0	0	0

Figure 20: Digitalized Survey



Educational Status

Educational Status	Qatari				Non-Qatari			
	Full time		Part time		Full time		Part time	
	Male	Female	Male	Female	Male	Female	Male	Female
Illiterate	0	0	0	0	0	0	0	0
Read & Write	0	0	0	0	0	0	0	0
Primary	0	0	0	0	0	0	0	0
Preparatory	0	0	0	0	0	0	0	0
High school	0	0	0	0	0	0	0	0
Pre.U. Diploma	0	0	0	0	0	0	0	0
Bachelor degree	0	0	0	0	0	0	0	0
Higher Diploma	0	0	0	0	0	0	0	0
M.A	0	0	0	0	0	0	0	0
Ph.D.	0	0	0	0	0	0	0	0

Figure 21: Digitalized Survey

Figure 22: Digitalized Survey

3. Database Setup

The database was automatically generated after the migration

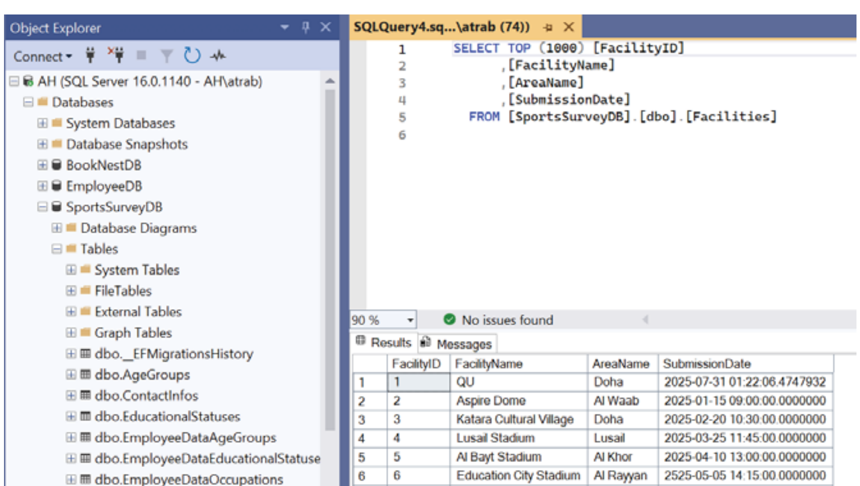


Figure 23: auto-generated database

4. Power Bi Dashboard

I used Bar charts, Cards, Slicers, Donuts, Buttons and Bookmarks to create a visualization for the data. It was made of 2 pages. In the beginning, I had to transform the data, unpivot some columns, create the fact tables, so that the visualization process is smooth.

Page 1: Contained a card that had the total employees from the AGE category of data, A bar chart that showed the total employees in the age data by age group range and origin whether Qatar or non Qatar, A dropdown slicer to filter the bar chart and a button to take you to the next page

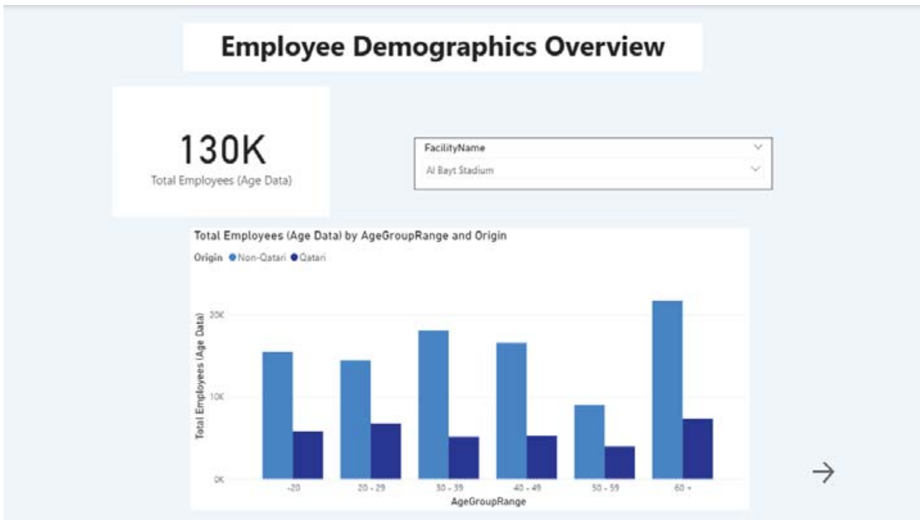


Figure 24: Page 1, Overview

Page 2: 3 Donuts that show the Total number of employees by Gender, by Origin and by EmployeeType. And a slicer to filter the donuts according to Occupation name.

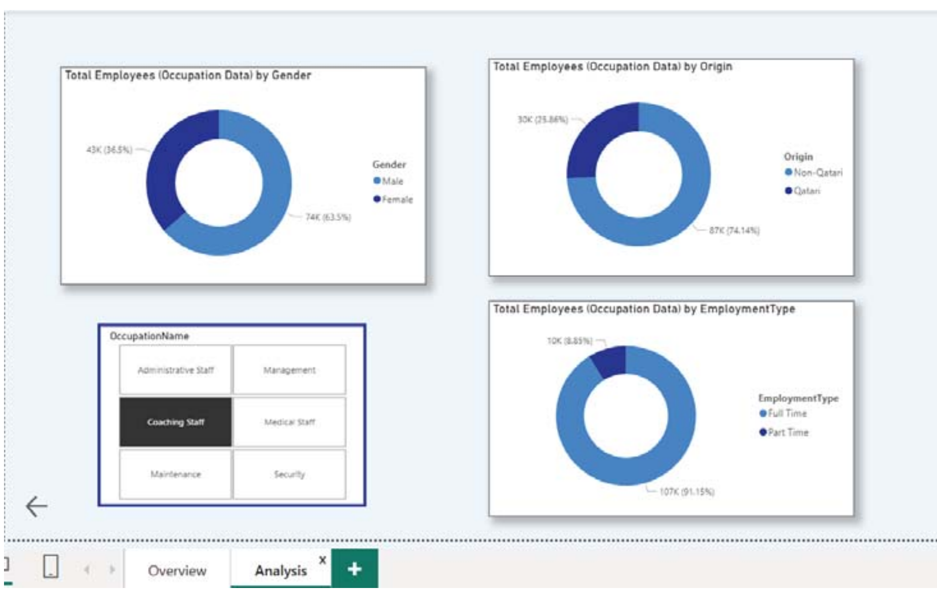


Figure 25: Page 2, Analysis

And in the second page I also included a Bookmark that shows the Occupation with most females, in this case was Administrative Staff.

Figure 26: Bookmark in page 2

And to achieve this I had to incorporate some **DAX measures** such as:

- Total Employees (Age Data) = SUM('Fact_EmployeeDataAgeGroups'[Count])
- Total Employees (Education Data) = SUM('Fact_EmployeeDataEducationalStatuses'[Count])
- Total Employees (Occupation Data) = SUM('Fact_EmployeeDataOccupations'[Count])

4. Reflections and learning outcomes

My internship experience was great, I made great connections, gained valuable insights and information. However, I would have liked to be working on an actual on-progress project within the organization, I believe that would have exposed me to real world projects, collaborations and skills. Despite that, I was grateful to have been given sessions on tools that were actually used within the organization. By the end of the internship, I was able to achieve my initial objectives from working on some Python exercises, creating interactive dashboards, learning how to utilize more than one technology and learning how to connect them such as Power Bi and SQL Server and many more. I also gained some new skills like got introduced to C# and .NET. I learned how to connect .NET and SQL Server to create a full stack project. I decided to go with .NET instead of Oracle in the Final project to challenge myself, however, in the future I would make sure I am more equipped to take the challenges and learn more background on my own by taking the initiative. Some academic courses that helped a lot with my internship experience is OOP, Web Development, Mobile Development, Data science and Database. The internship also made me consider Data science as a future career, I enjoyed the especially the data visualization part.